

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

Name: Pravin R

Reg. No: 192421292

Date: 11/08/2026

Experiment 13: Car Price Prediction Model using Python

Aim

To implement a Car Price Prediction model using Linear Regression in Python and predict the price of a car based on its features.

Procedure

1. Import the required libraries.
2. Create a car dataset containing features such as year, mileage, engine size, and car age.
3. Separate input features and target price.
4. Split the dataset into training and testing data.
5. Train the Linear Regression model.
6. Predict car prices using the test data.
7. Evaluate the model using MSE and R^2 score.
8. Predict the price of a new car.

Algorithm

1. Start.
2. Load the car dataset.
3. Select the input features and target variable.
4. Split the dataset into training and testing sets.
5. Train the Linear Regression model.
6. Predict prices for the test data.
7. Calculate MSE and R^2 score.
8. Use the trained model to predict the price of a new car.
9. Display the results.

Dataset 1 — Basic Car Features

Age	Mileage	Engine_CC	Horsepower	Price
2	15000	1200	85	850000
3	25000	1500	100	750000
4	40000	1600	110	680000
5	55000	1800	130	600000
7	70000	2000	150	500000

Dataset 2 — Sedan Price

Car_Age	Kilometers	Engine_Size	Fuel_Efficiency	Price
1	10000	1200	20	950000
2	18000	1500	18	850000
3	30000	1500	17	780000
5	45000	1800	15	650000
7	65000	2000	13	520000

Dataset 3 — Used Car Price

Year	Mileage	Engine_CC	Owners	Price
2023	8000	1200	1	900000
2022	15000	1500	1	820000
2020	30000	1500	1	700000
2019	45000	1600	2	600000
2017	65000	1800	2	480000

Dataset 4 — SUV Price

Age	Mileage	Engine_CC	Horsepower	Price
1	12000	1500	110	1200000
2	20000	1800	140	1100000
3	35000	2000	160	980000
5	50000	2200	180	820000
7	75000	2500	200	650000

Dataset 5 — Premium Car Price

Age	Mileage	Engine_CC	Horsepower	Price
1	5000	2000	180	2500000
2	12000	2200	200	2200000
3	20000	2500	230	1950000
4	35000	2800	260	1700000
6	50000	3000	280	1400000

Code

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([
    [2018, 25000, 1.2],
    [2019, 20000, 1.5],
    [2017, 35000, 1.0],
    [2020, 15000, 1.5],
    [2021, 10000, 2.0],
    [2016, 50000, 1.2],
    [2019, 30000, 1.5],
    [2022, 8000, 2.0],
    [2015, 60000, 1.0],
    [2020, 18000, 1.8],
    [2018, 40000, 1.5],
    [2021, 12000, 2.0]
])

y = np.array([
    750000, 850000, 600000, 1050000,
    1300000, 500000, 780000, 1450000,
    400000, 1100000, 650000, 1250000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Prices:")
print(y_test)

print("Predicted Prices:")
print(y_pred)

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))

new_car = [[2022, 10000, 1.5]]

prediction = model.predict(new_car)

print("Predicted Car Price:", prediction[0])
```

Output

```
... Actual Prices:
[ 650000 1100000  750000]
Predicted Prices:
[ 807268.60299599 1103938.52165797  723951.99209384]
Mean Squared Error: 8475000038.454495
R2 Score: 0.7722917243400285
Predicted Car Price: 1247725.054045409
```

Result

The Car Price Prediction Model was successfully implemented using Linear Regression. The model predicts the price of a car based on its manufacturing year, mileage, and engine size and evaluates its performance using MSE and R^2 score.

